

Economic History II

Media, propaganda and opinion

FGV EPGE

Outline of the class

- **DellaVigna, Enikolopov, Mironova, Petrova and Zhuravskaya (2014)**, the paper you read: Serbian radio spilling across the border into Croatia.
- Adena, Enikolopov, Petrova, Santarosa and Zhuravskaya (2015): radio in Weimar and Nazi Germany.
- Bursztyn and Cantoni (2016): West German television in East Germany.
- Same tool in all three: signal propagation and terrain decide who hears what.
- Different regimes, different content, different listeners.

Serbian radio in Croatia

DellaVigna, Enikolopov, Mironova,
Petrova and Zhuravskaya (2014)

Cross-border media and nationalism: Evidence from Serbian radio in Croatia ¹

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Abstract

How do nationalistic media affect animosity between ethnic groups? We consider one of Europe's deadliest conflicts since WWII: the Serbo-Croatian conflict. We show that, after a decade of peace, cross-border nationalistic Serbian radio triggers ethnic hatred towards Serbs in Croatia. Mostly attracted by non-political content, many Croats listen to Serbian public radio (intended for Serbs in Serbia) whenever signal is available. As a result, the vote for extreme nationalist parties is higher, and ethnically offensive graffiti are more common, in Croatian villages with Serbian radio reception. A laboratory experiment confirms that Serbian radio exposure causes anti-Serbian sentiment among Croats.

The question

- Which factors fuel animosity between ethnic groups?
- Here: media meant to strengthen the identity of one group, heard by a rival group.
- Radio and television waves travel across political and ethnic boundaries.
- Common where neighbors share a language: the former Yugoslavia, the former USSR, Sudan.
- Also within countries: Muslims and Hindus in India, Basques and Spanish in Spain.

Three answers are possible a priori

- **Animosity rises**: views polarize, future conflict becomes more likely.
- Or exposure reduces informational asymmetries between groups and eases tension.
- Or the cross-group effect is negligible: people mostly consume media that support their own views.
- The literature had studied propaganda on its *intended* audience. Not on the audience across the border.

The conflict

- Croatian-Serbian conflict, 1991 to 1995, part of the Yugoslavian wars.
- Croats predominantly Catholic, Serbs predominantly Orthodox. Animosity escalating since the 1980s.
- Massacres of civilians and ethnic cleansing on both sides.
- Roughly 250,000 displaced civilians and 20,000 casualties, the vast majority civilians.
- Milosevic overthrown in 2000 and handed to the Hague International Criminal Tribunal.

The region

- Eastern Slavonia, Baranja and Western Srijem, on the border with Serbia.
- Site of full-scale armed conflict in 1991, under Serbian occupation until 1995.
- Two counties. Vukovar-Sirmium suffered most; all Croats there were displaced or killed during the Krajina period.
- Baranja has a large Hungarian minority and much weaker Croatian nationalism.
- The comparison is always **within county**.

Why the setting works

- “The Serb-Croat case is a nearly ideal setting to study cross-group effects of media.”
- Serbs and Croats speak the same language, with different alphabets.
- The gap is about the size of the gap between British and American English.
- Serbian public radio is made for listeners inside Serbia, and its signal reaches some Croatian villages and not others.

Why radio, and not television or newspapers

- Rural villages: low penetration of television sets and newspapers.
- Radio is cheap and easily accessible, the main political information source since Socialist Yugoslavia.
- A typical Croatian listens to radio about 250 minutes a day.
- Most people leave the radio on all day, and many listen on the street.
- Local stations dominate: the three national public channels hold a joint share of just 10% of radio audiences.

Serbian public radio

- Four RTS public stations: news, art, intellectual programs, and Radio Belgrade 202 (popular music).
- They “operate with the official mission to strengthen Serbian national identity”.
- 9% of coverage goes to neighboring countries, focused on war and war crimes there.
- **Turbo-folk**, invented under Milosevic, is considered offensive by many Croats and Bosnians: it glorifies Serbs during the war.
- One private exception: Radio B92, founded in 1989 by the Soros foundation and USAID.

Croats are an off-target audience

- The vast majority of listeners to these stations are Serbs in Serbia.
- If the target were Croats, the transmitters would sit closer to the border to push the signal in.
- So the paper identifies “what is likely to be an unintended media effect on an off-target audience as a catalyst of ethnic hatred”.

Two measures of reception

- **Hand-collected**: the authors drove through the region in June 2009 and June 2010 with a portable receiver.
- In the center of each village, near the church or the war monument, they tested whether RTS stations came in.
- Available if two people judged the quality good enough to listen to. 139 villages, 16 with reception.
- **Predicted signal strength**: RTS transmitter locations and power plus topography, through the Irregular Terrain Model.
- 417 villages within 75 kilometers of the border.

The two measures are complementary

- The hand measure captures fine variation the signal model cannot: forests, obstacles off the line of sight.
- It also carries unavoidable measurement error, and depends on the receiver used.
- The signal model covers a much larger sample, but ignores non-topographic obstacles.
- The two are positively correlated, and the overlap is far from complete.

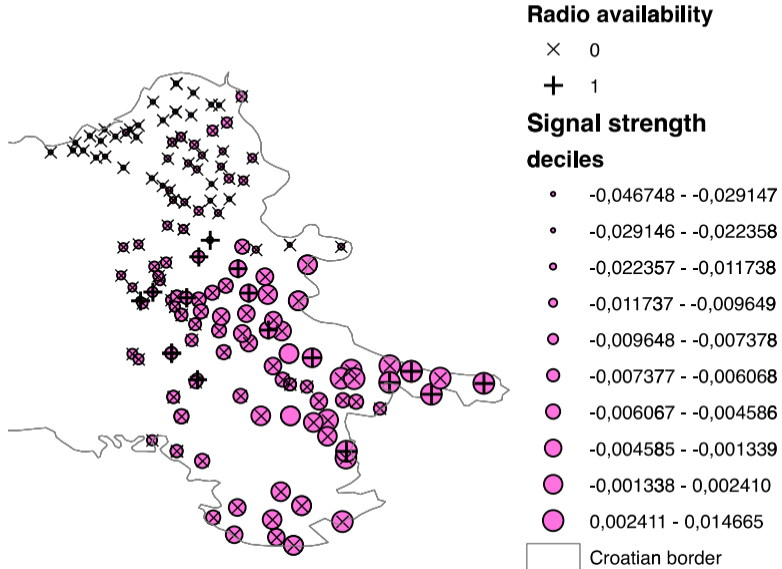
Figure 1a. Map of the area with the baseline and extended samples of villages



Reading the map

- Hollow circles: the 139 villages visited in person (baseline sample).
- Solid dots: the 417 villages within 75 km of the border (extended sample).
- Triangles: the Serbian transmitters, well inside Serbia.
- Vukovar and Osijek are excluded: cities, not comparable to the rural villages.

Figure 1b. Map of the area with the baseline sample of villages showing both the measured and the predicted reception of Serbian radio.



Measured reception against predicted signal

- Crosses and plus signs: RTS unavailable and available on the visit.
- Circle size: predicted signal strength, in deciles.
- Availability declines with distance from the Serbian border, but in a jagged way.
- **The jaggedness is the variation the paper uses.**

What is measured on the Croatian side

- Votes for HSP and its two splinters, HP-HPP and HCSP: the extreme nationalist bloc.
- HDZ, the main center-right party, is moderately nationalist. SDP is the neutral one, most popular among the Serbian minority.
- Ethnically offensive graffiti in public spaces in the village center: 36 of the 139 villages.
- Turnout, from the Central Election Commission, aggregated from polling stations to villages.

The parties

- HDZ took 36.6% of the national vote in 2007 and 66 seats; SDP 31.2% and 56 seats.
- HSP is the main extreme nationalist party, with the ideology of “the Great Croatia”.
- In the border region it is not marginal: 8% of the vote in the north, 5.2% in the south.
- In the 1990s HSP ran its own paramilitary unit, with symbols resembling those of the wartime fascist state.

Why the 2007 election

- Third post-war parliamentary election, and the first in a relatively stable economic and political environment.
- In 2003, many refugees from the region had still not returned home.
- In 2011, the last election before accession to the European Union: nationalistic issues were deliberately kept out of the campaign.
- Reception was measured in 2009 and 2010; transmitters were unchanged from 2007 to 2011.

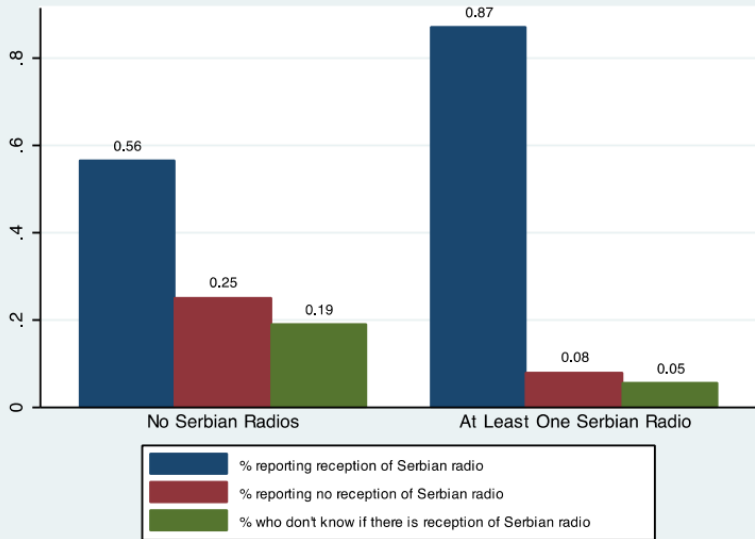
First question

Do Croats actually listen?

The street survey

- December 2010 and January 2011: 70 people interviewed in 9 villages.
- 4 villages coded as having no Serbian radio, 5 with at least one station.
- Interviewers stopped people on the central street. Response rate about 50%.
- Small and not representative, so the paper treats it as suggestive.

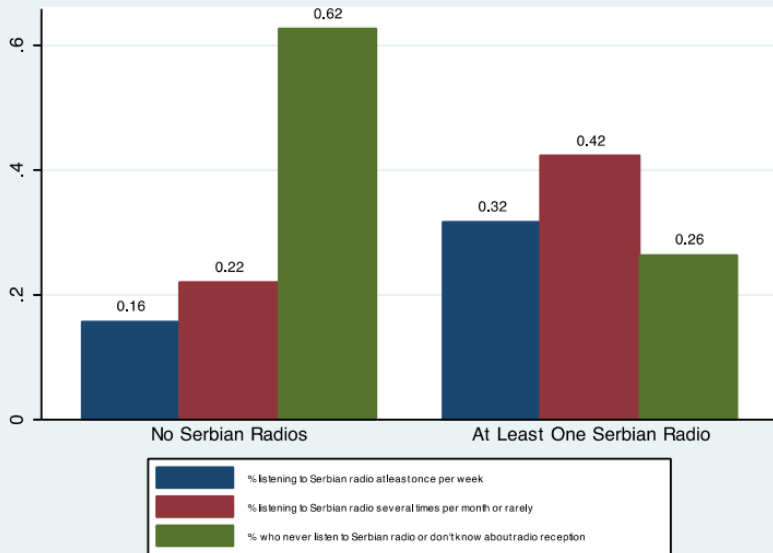
Figure 2a. Reported reception of Serbian radio (survey response), as function of availability of Serbian radio in village.



Reported reception

- Villages coded as having at least one Serbian station: 87 percent say they receive it.
- Villages coded as having none: 56 percent still say they do.
- Not a contradiction: better receivers and amplifiers are common, people listen in cars and in neighboring villages, the weather moves the signal.
- The hand measure is a floor, not a perfect classification.

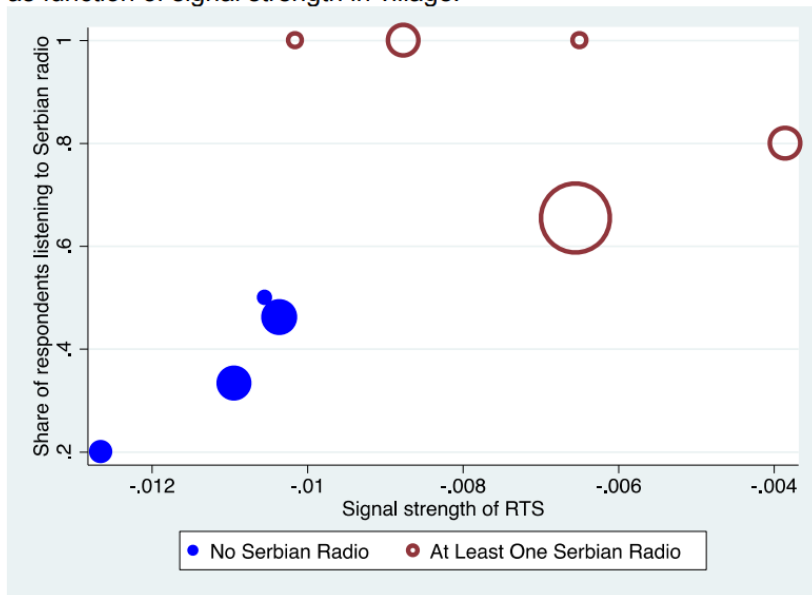
Figure 2b. Reported frequency of listening to Serbian radio (survey response), as function of availability of Serbian radio in village



How often they listen

- Villages with at least one Serbian station: 26 percent never listen, 32 percent listen at least once a week.
- Villages without: 62 percent never listen, 16 percent at least once a week.
- In areas where the signal is available for ordinary radio sets, **78% acknowledge listening occasionally**.
- That sits badly with the extreme view that people only consume media that match their politics.

Figure 2c. Reported incidence of listening to Serbian radio (survey response), as function of signal strength in village.



The same pattern with predicted signal

- Horizontal axis: predicted RTS signal strength in the village.
- Vertical axis: share of respondents who report listening to Serbian radio.
- Marker size is proportional to the number of respondents in that village.
- Both reception measures line up with what people say they do.

Why would a Croat tune in?

- Respondents who report listening are also more likely to support concerts by Serbian artists and to know their new songs.
- No such correlation for reported listening to Croatian or Bosnian radio.
- A number of former Socialist Yugoslavian singers, popular among Croats, are banned from Croatia for political reasons.
- “a preference for the entertainment content embedded in the RTS programming may be the main reason for the high exposure to Serbian radio, despite its nationalistic content”.

Substitution

- No difference across village types in reception of, or listening to, Bosnian radio.
- Croatian radio reaches everyone: every respondent reports reception.
- Where Serbian radio is available, people listen to Croatian radio less *frequently*.
- So the Serbian signal partly displaces Croatian listening, but does not replace it.

Second question

Is the content actually inflammatory?

A laboratory experiment in Vukovar

- Croatian students from the region, held at the university of Vukovar.
- Three groups, each hearing a short remix: three news segments separated by two songs.
- The first two news segments are the same for everyone, taken from a Croatian station.
- What differs: the two songs and the third news segment.

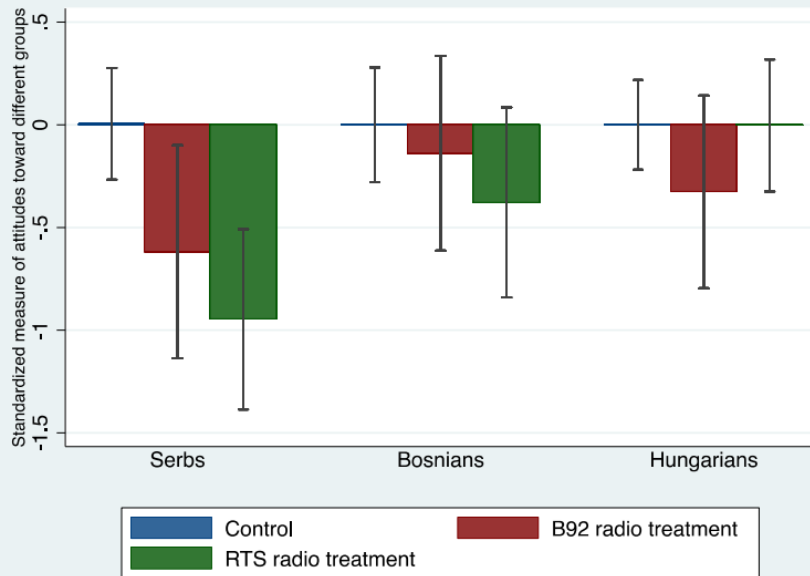
The three arms

- **Control**: Croatian radio, two popular Croatian songs.
- **B92**: the independent Serbian station, two popular Serbian songs.
- **RTS**: the Serbian public station, substantially more nationalistic, another two Serbian songs.
- The Serbian arms carry the same news story, on the Dick Marty report about Kosovo.
- None of the news broadcasts mentions the Serbo-Croatian conflict.

The music, and what is asked afterwards

- Both Serbian arms use modern songs with direct references to Serbian national folklore, easily identifiable by Croats.
- One song in the RTS arm is devoted to a former Serbian paramilitary commander accused of war crimes in Croatia.
- After listening, subjects answer questions about Serbs, Bosnians, Hungarians and Rusini.
- Willingness to work with them, to send children to school with them, to have them as friends, as in-laws, plus a feeling thermometer.

Figure 3. Attitudes toward different ethnic groups (Laboratory Experiment)



Reading the experiment

- Both Serbian treatments push attitudes toward **Serbs** down. The RTS arm pushes them further than B92.
- No movement in attitudes toward Bosnians or Hungarians, as expected.
- B92 is neutral toward Croats; RTS is openly hostile.
- So the channel is not just being reminded of the former war enemy. It is the nationalistic content itself.

Third question

Does it change behavior in the villages?

Is reception as good as random?

- Of 15 non-geographic village characteristics, only education predicts the hand-measured availability.
- Taken together, those characteristics do not predict it. Consistent with reception being idiosyncratic.
- Predicted signal strength is different: it is related to distance to Serbia, population, and the share disabled in the war.
- So for the signal measure, controls and robustness checks carry more weight.

What if something unobserved drives both?

- If an omitted village trait inflated the result, adding controls should pull the estimate down.
- It does not: the estimate is essentially identical with and without controls.
- The observable village traits that predict radio availability are, if anything, *negatively* related to nationalist voting.
- Under the usual assumption that the unobserved looks like the observed, the estimates are biased **against** the result.

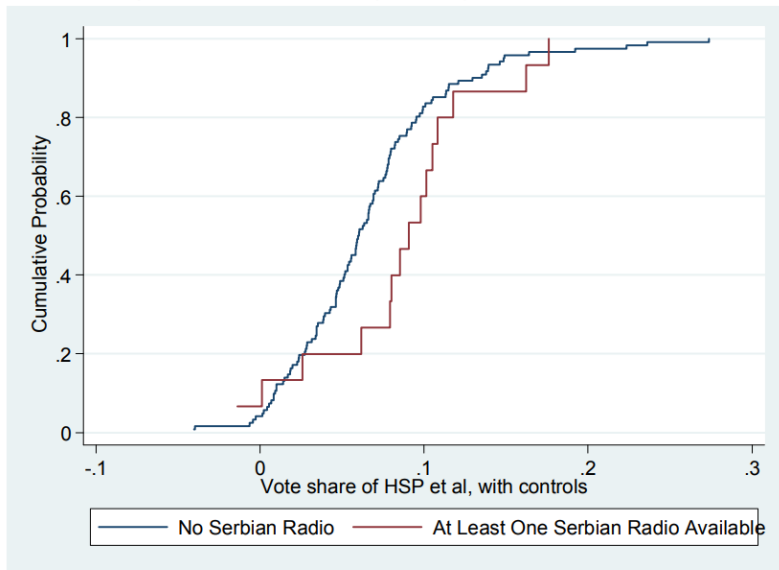
The election result

- Villages with at least one RTS station available: **2.6 percentage points** more vote for the extreme nationalist parties.
- Against a baseline vote share of 7 percentage points in the region.
- So a substantial part of the ultra-nationalist vote in the border region is explained by Serbian radio reception.
- The predicted-signal measure gives the same answer, and adding controls barely moves it.

One radio or several

- 9 of the 139 villages receive more than one RTS station.
- There the audience is more likely to run into Serbian content, deliberately or not, and reception is better.
- Those villages show a larger effect on nationalist voting, though the difference between one and several is not sharp.

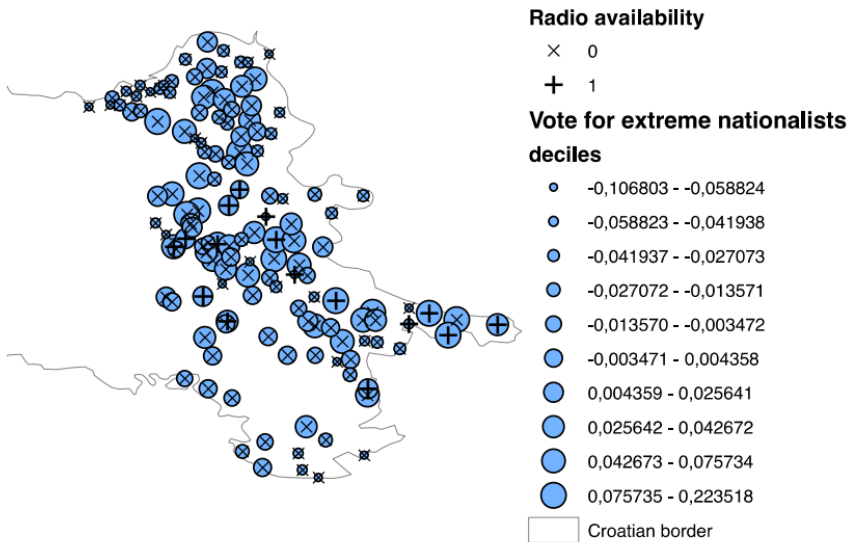
Figure 4a. Vote share for extreme nationalistic parties in villages with, and without, reception of Serbian radio (controls).



Not one outlier village

- Cumulative distribution of the nationalist vote share, after taking controls into account.
- Villages with Serbian radio nearly dominate villages without: the whole distribution shifts right.
- A formal test rejects that the two distributions are the same.
- With only 16 treated villages, this matters: the result is a shift, not a single village.

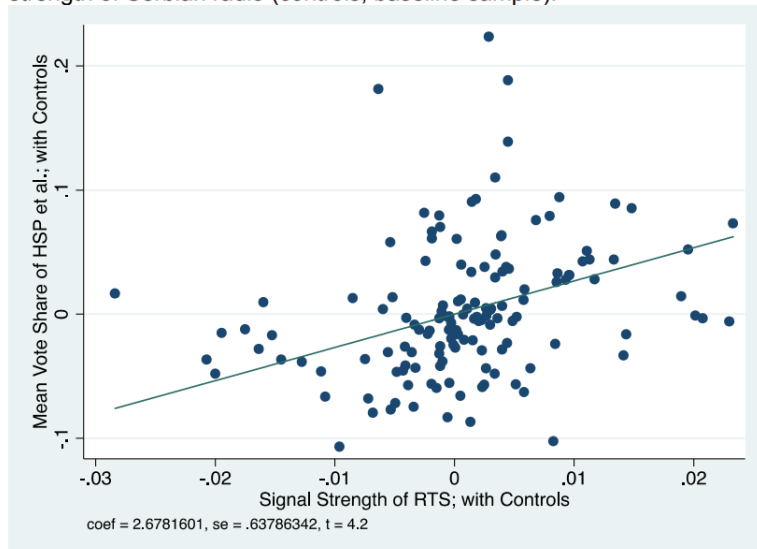
Figure 4b. Map of the vote share for extremely nationalistic parties (with controls) and the measured reception of Serbian radio.



The same thing on the map

- Plus signs: villages where RTS was received. Crosses: villages where it was not.
- Circle size: the nationalist vote share, in deciles, after controls.
- Big circles cluster on the plus signs, along the eastern edge.

Figure 5. Vote share of extremely nationalistic parties as function of signal strength of Serbian radio (controls, baseline sample).



Notes: The picture shows share of votes for HSP et al. and predicted signal strength of Serbian radio after taking controls into account. Residuals are taken from the specification presented in column (3) of Table 3.

With the continuous measure

- Each dot is a village: predicted signal strength against nationalist vote share, both after controls.
- The relationship is monotonic across the range, not driven by the tails.
- Same conclusion as the on/off measure, on a sample that does not depend on the field visits.

The other parties

- The moderate nationalist party (HDZ) loses vote share where Serbian radio is available.
- Some evidence that the social-democrats (SDP) gain.
- Reading: marginal voters shift from moderate to extreme nationalism, and the electorate polarizes.
- No evidence of an effect on turnout, though the count of listed voters is noisy.

A non-electoral measure: graffiti

- Graffiti coded following the method of the Vukovar NGO Globalpact.
- Examples: “Ubi Srbina” (“Kill a Serb”) and “Srbe na vrbe” (“Hang a Serb on a willow”).
- In villages with Serbian radio, the probability of ethnically offensive graffiti is **35 to 40 percentage points** higher.
- That is roughly double the rate in villages without it.

How large is the effect, per listener?

- *Persuasion rate*: the fraction of the audience convinced to change behavior by exposure to the outlet.
- Inputs: exposure to Serbian radio is 31 percentage points higher in villages with a station; turnout is 56.2%; the vote effect is 2.6 pp.
- Using the on/off measure: **4.3%**. Using predicted signal strength: **2.7%**.
- On the low end of the persuasion rates the media literature has estimated.

A radio that should do nothing

- Hungarian radio also crosses into the region, and very few Croats speak Hungarian.
- It has no effect on nationalist voting, and controlling for it leaves the Serbian effect intact.
- Controlling for the signal of the two major Croatian networks changes nothing either.
- So the result is not about villages that simply happen to get good radio reception in general.

Other checks

- Allowing for the fact that neighboring villages are not independent does not weaken the result.
- Controlling for driving time and distance to the border, and for war experience, leaves it.
- Controlling for free-space signal (what the signal would be with a clear line of sight): the hand measure no longer stands out, the continuous measure does, less precisely.
- Comparing each village with reception to its 5 most similar villages without: the same answer.

Other elections

- 2011: same pattern, exposure to Serbian radio raises the extreme nationalist vote.
- 2003: no evidence in the baseline sample, some in the extended sample.
- Why 2003 differs: transmitters were still being repaired, and many people still lived and voted in refugee camps.
- Reception was measured in 2009 and 2010, so 2003 is the furthest extrapolation.

Where the effect is smaller

- The effect shrinks in villages with more people disabled in the war of independence.
- It also shrinks in villages that played an important role in the war.
- There nationalism is already high from lived memory, so the radio message is infra-marginal.
- Media move the people who are not already convinced.

What the paper documents

- A decade after the shooting stopped, Serbian public radio still crossed the border and was heard by Croats.
- Entertainment pulled them in; political content came with it.
- Where it was received: more ultra-nationalist voting, more offensive graffiti, a more polarized electorate.
- “nation-building efforts implicit in the nationalistic content of the group-specific media (in our case, the Serbian radio) can have important negative spillovers on the persistence of peace.”

Radio and the rise of the Nazis

Adena, Enikolopov, Petrova,
Santarosa and Zhuravskaya (2015)

The question

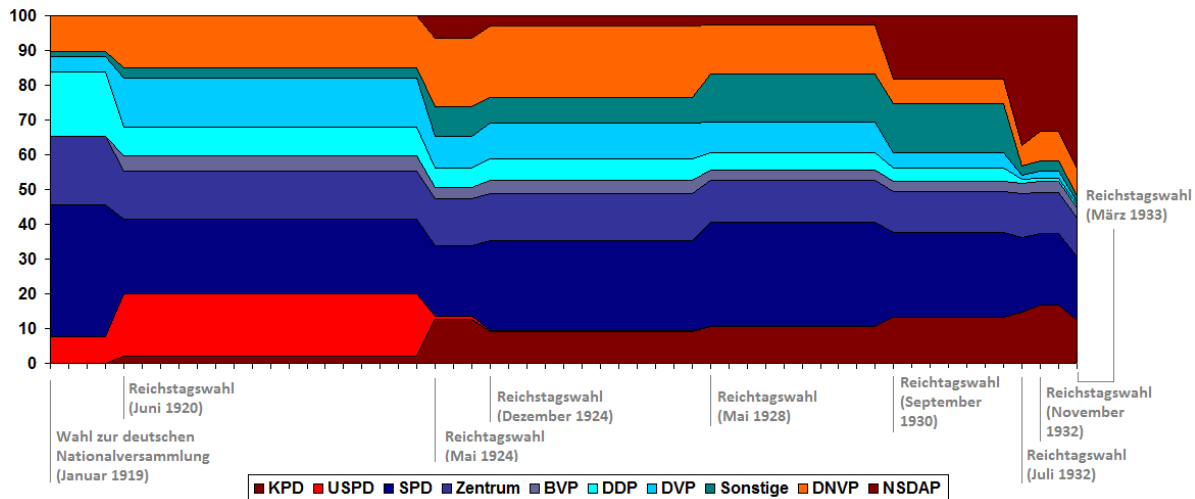
- How do the media affect public support for democratic institutions in a **fragile democracy**?
- What role do they play once the dictatorship is in place?
- The rise of the Third Reich was the most prominent collapse of a democracy without a military coup.
- Same team of authors, overlapping with the Croatia paper, same signal-propagation toolkit.

Goebbels believed it worked

- At the Berlin radio exhibition, August 1933: “It would not have been possible for us to take power or to use it in the ways we have without the radio.”
- In his diary, on the weeks before the last competitive election: “Now it will be easy to carry on the fight, for we can call on all the resources of the State. Radio and press are at our disposal. We shall stage a masterpiece of propaganda”.
- Historians disagree. Some read the claims as self-serving, and argue propaganda added little.
- Nobody had tested it systematically before this paper.

Weimar politics

- Parliamentary democracy from 1919, governed by a coalition of centrist parties led by the SPD.
- It survived hyperinflation and held a stable majority until 1930.
- The Depression, the recall of US short-term loans and the reparations burden broke it.
- NSDAP vote: 2.6% in 1928, 18.3% in 1930, 37.3% in July 1932, 33.1% in November 1932.
- Hitler appointed chancellor on January 30, 1933; the NSDAP then took 43.9% in March.



Reading the Reichstag chart

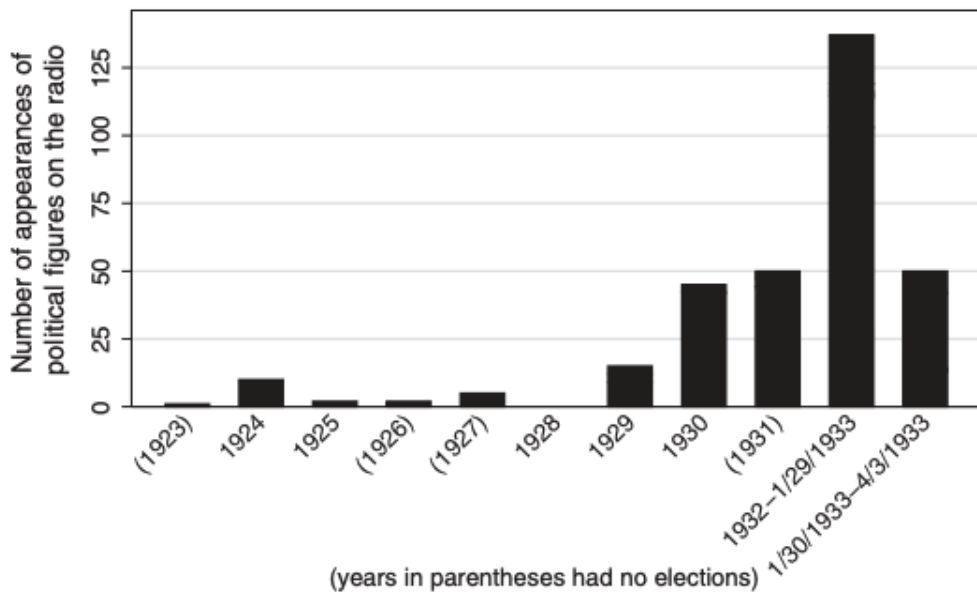
- Vote shares by party, from the 1919 National Assembly election to March 1933.
- The dark red band at the top is the NSDAP: negligible until 1928, then it swallows the right.
- The SPD (dark blue) erodes slowly; the KPD (dark red at the bottom) grows through the Depression.
- The center collapses from both ends at once.

Radio content changed twice

- Until 1928: deliberately apolitical. A 1926 regulation forbade partisan content; no campaign content aired in 1928.
- **1929**: right-wing parties organize a referendum to repudiate the reparations of the Treaty of Versailles.
- The government answers with an intensive campaign against it. Radio turns pro-government and pro-democratic.
- The Nazis and the communists are denied airtime, unlike the other parties.
- **January 30, 1933**: Hitler takes the executive powers, radio among them. Content turns pro-Nazi.

A

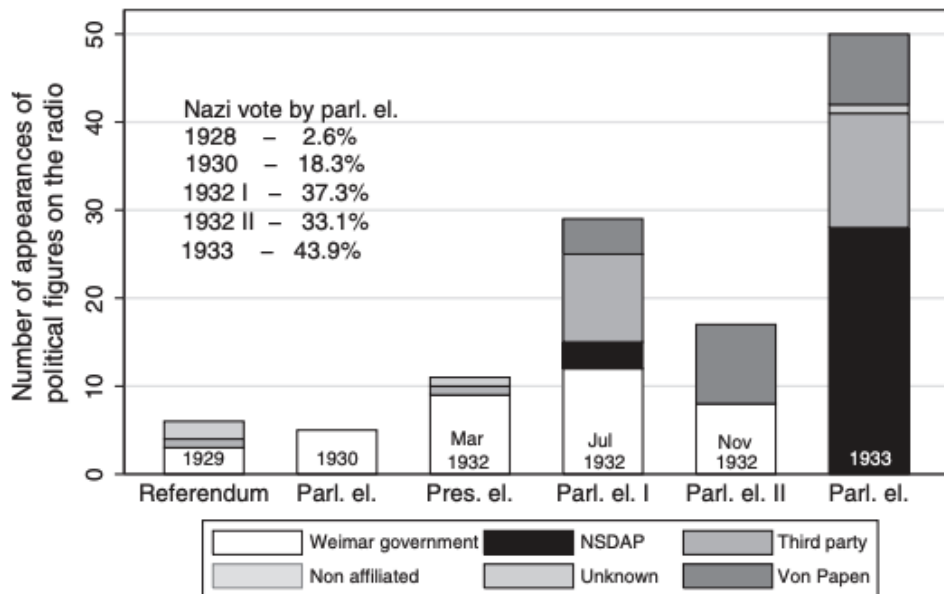
Political broadcast by year



Political figures on the air, by year

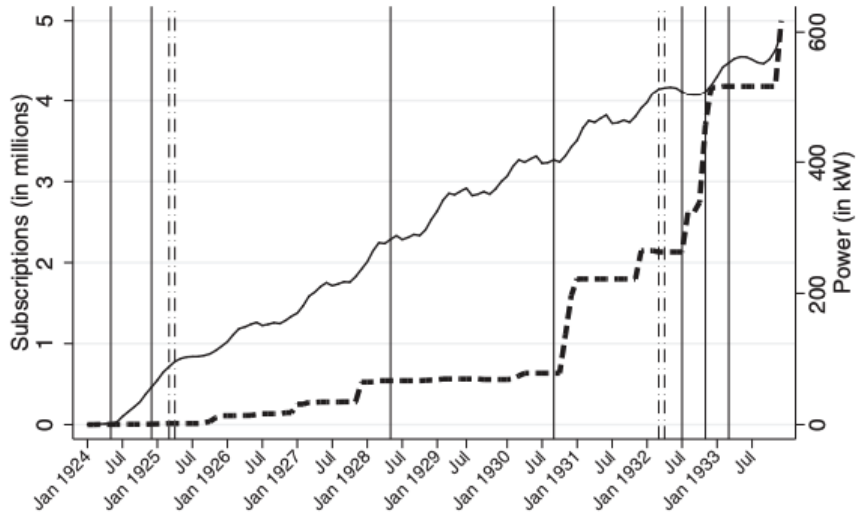
- Counts of radio appearances by government officials, party representatives and members of parliament.
- Almost nothing between 1925 and 1928, in years with and without elections.
- A discontinuous jump in 1929, then a rising trend.
- The peak bar is 1932 to January 29, 1933, just before Hitler takes over the transmitters.

B Political broadcast by election campaign and affiliation



Who got the airtime

- Same counts, now split by campaign and by party affiliation.
- Through 1932 the bars are Weimar government and third parties. The black NSDAP block is absent.
- In the 1933 campaign the black block dominates the tallest bar.
- The Nazi vote share for each election is printed in the corner: 2.6, 18.3, 37.3, 33.1, 43.9.



— Radio Subscriptions (in millions)

— Power (in kW)

— parliamentary elections date

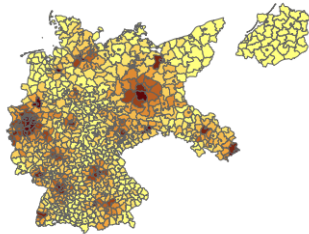
— presidential election date

Radio reach over the same decade

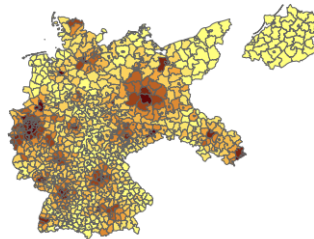
- Solid line: radio subscriptions, from near zero in 1924 to above 4 million by 1932.
- Dashed line: total transmitter power in kilowatts, which jumps in steps as new transmitters open.
- Vertical lines mark parliamentary and presidential elections.
- Both the audience and the map of who can hear are moving during the period studied.

Figure A1. Radio Signal Strength in Germany. Raw signal strength measure.

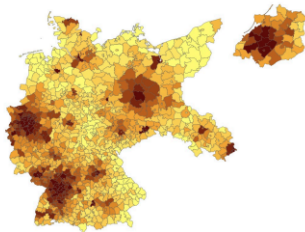
A. May 1928



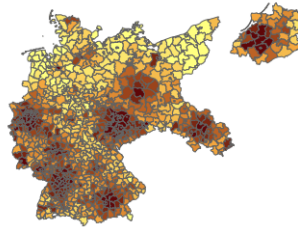
B. September 1930



C. July 1932



D. March 1933



Where the signal reached

- Predicted signal strength by district, in May 1928, September 1930, July 1932 and March 1933.
- Darker is stronger. Coverage thickens around the new high-power transmitters.
- Computed from transmitter location and power plus terrain, the same Irregular Terrain Model as in Croatia.
- **The variation is over time and space at once:** which districts could hear, and when.

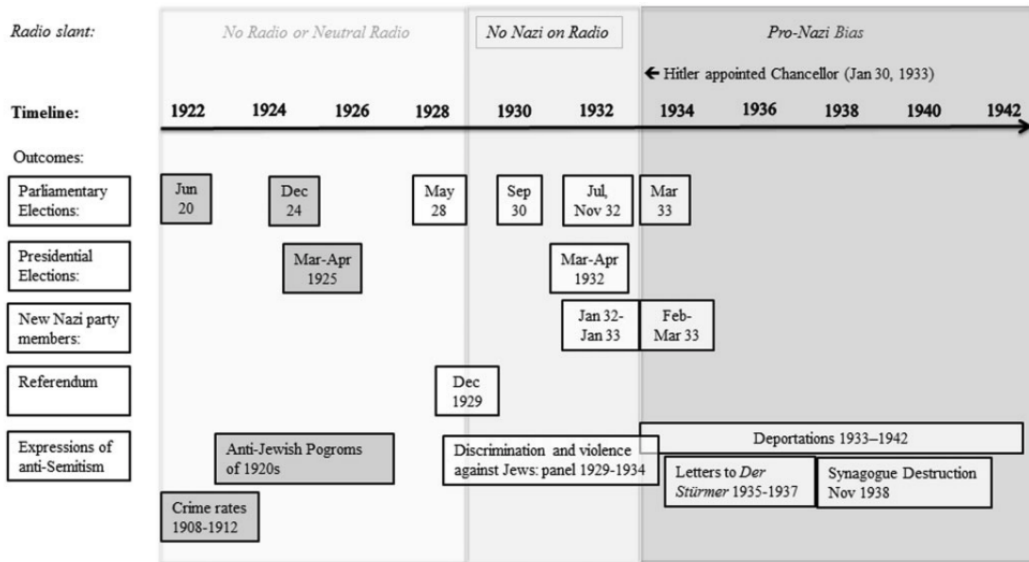


FIGURE III

The Timing of the Measurement of Outcome Variables and of the Change in Radio Content

What is measured, and when

- Top band: the slant of the radio. No radio or neutral, then no Nazi on radio, then pro-Nazi from 1933.
- Outcomes below: parliamentary and presidential elections, new Nazi party members, the 1929 referendum.
- Expressions of antisemitism: discrimination and violence 1929 to 1934, deportations 1933 to 1942, letters to a Nazi newspaper, synagogue destruction in November 1938.
- Crime rates from 1908 to 1912 and the pogroms of the 1920s sit before the radio: they are the checks that should find nothing.

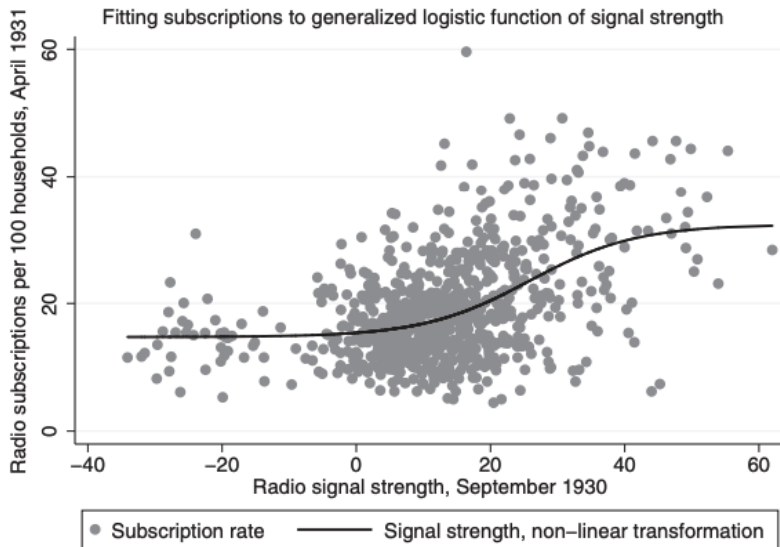


FIGURE IV

Radio Subscriptions and Signal Strength, 1931 (No Controls)

From signal to listeners

- Each dot is a district: predicted signal strength against actual radio subscriptions per 100 households.
- The fitted S-curve is flat at both ends and steep in the middle, which is how radio reception behaves.
- Below a threshold there is only noise; above one, extra strength adds nothing.
- The paper uses both the raw signal and this transformation of it.

TABLE I
RADIO SUBSCRIPTIONS AND SIGNAL STRENGTH

	(1)	(2)	(3)	(4)
	Share of households with radio subscription at a given date			
Date for the subscription rate	Apr 1931	Apr 1932	Apr 1932	Apr 1933
Date for the signal strength	Sep 1930	Jul 1932	Nov 1932	Mar 1933
Panel A				
Radio signal strength	0.218*** [0.030]	0.175*** [0.028]	0.161*** [0.031]	0.163*** [0.036]
Distance to the nearest big city, city dummy, altitude	Yes	Yes	Yes	Yes
Socioeconomic controls	Yes	Yes	Yes	Yes
Population, fifth-order polynomial	Yes	Yes	Yes	Yes
Voting controls, 1924	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes
Observations	858	883	883	885
R^2	0.576	0.531	0.526	0.514
F -stat for the signal strength variable	50.19	36.13	24.53	20.30
Panel B				
Radio signal strength, nonlinear transformation	0.671*** [0.096]	0.683*** [0.100]	0.656*** [0.120]	0.602*** [0.123]
Distance to the nearest big city, city dummy, altitude	Yes	Yes	Yes	Yes
Socioeconomic controls	Yes	Yes	Yes	Yes
Population, fifth-order polynomial	Yes	Yes	Yes	Yes
Voting controls, 1924	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes
Observations	858	883	883	885
R^2	0.579	0.533	0.525	0.513
F -stat for the signal strength variable	46.62	42.41	26.38	23.05

Signal predicts subscriptions

- Four subscription dates between April 1931 and April 1933, each paired with a nearby signal map.
- The relationship is strong and stable across all four.
- This is what licenses using signal strength as a stand-in for exposure where subscription data do not exist.

TABLE III
RADIO AVAILABILITY AND VOTING FOR THE NAZIS: DISTRICT FIXED EFFECTS

	(1)	(2)	(3)	(4)	(5)	(6)
	Nazi vote share				Change in the Nazi vote share	
Time span of the sample	September 1930, July 1932, and November 1932		All parliamentary elections 1928–1933, combined		Between September 1930 and November 1932	
Specification	Panel: OLS	Panel: OLS	Panel: OLS	Panel: OLS	First differences: OLS	First differences: IV
Radio signal strength	−0.079** [0.034]					
Radio signal strength, nonlinear transformation		−0.227** [0.088]				
Radio signal strength × pro-Nazi slant			0.028*** [0.007]			
Nonlinear transformation of signal strength × pro-Nazi slant				0.123*** [0.027]		
Change in subscription rate between April 1931 and April 1932					−0.104 [0.098]	−2.127** [0.873]
Baseline controls, interacted with time fixed effects	Yes	Yes	Yes	Yes	No	No
Baseline controls	No	No	No	No	Yes	Yes
District fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	No	No
Observations	2,836	2,836	4,713	4,713	827	827
Number of districts	959	959	959	959	827	827
R^2	0.939	0.939	0.972	0.972	0.658	
F-statistics for instrumental variable						26.44

Notes. Standard errors clustered at the region level in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$. Pro-Nazi slant equals 0 in 1928, −1 between 1930 and 1932, +1 in 1933. Controls include fifth-order polynomial of population, share of Jewish and Catholic population, shares of blue- and white-collar workers in 1925, city dummy, number of war participants per 1,000, number of welfare recipients per 1,000, number of renters of social housing per 1,000, log of average property tax, altitude, share of unemployed and partially employed, distance to the nearest city with population over 50k, turnout and vote shares of DNVP, NSFP, SPD, and Zentrum in December 1924. Number of observations changes between elections because of redistricting. In column (8) change in radio subscription rate is instrumented using change in nonlinear transformation of the signal strength.

The main comparison

- Columns 1 and 2: between 1930 and 1932, when the radio was pro-Weimar, more signal means *fewer* Nazi votes.
- Columns 3 and 4: pooling all five elections from 1928 to 1933 and letting the effect flip with the slant of the content.
- Slant is coded 0 in 1928, -1 from 1930 to 1932, $+1$ in 1933.
- Columns 5 and 6 use the change in actual subscriptions between April 1931 and April 1932.
- Each district is compared with itself over time, so fixed district traits cannot drive it.

Pro-Weimar radio slowed the Nazis

- 8.2 percentage points more predicted radio subscription in 1930 (one standard deviation of the actual rate) → a **1.8 pp smaller** rise in the Nazi vote between 1928 and 1930.
- Using actual subscriptions rather than predicted ones, the same exercise gives 2.9 pp.
- Tracking the change in subscriptions district by district between September 1930 and November 1932: a fall of 4.9 pp.
- Counterfactual with no radio during the September 1930 campaign: the Nazis get 22.3% instead of 18.2%.
- The SPD got 24.5%. That gap mattered for bargaining over the chancellorship already in 1930.

TABLE IV
RADIO AND AN INCREASE IN NAZI VOTE SHARE

	(1)	(2)	(3)	(4)
	Change in vote share of the Nazi party since previous elections			
Election dates	Sep 1930 (change from May 1928)	Mar 1933 (change from Nov 1932)		
Panel A: reduced-form estimation				
Radio signal strength	−0.061*** [0.022]		0.045** [0.020]	
Radio signal strength, nonlinear transformation		−0.217*** [0.071]		0.128* [0.071]
Region fixed effects	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes
Observations	958	958	918	918
Panel B: OLS and IV results				
Date for the subscription rate variable	Apr 1931		Apr 1933	
Specification	OLS	IV	OLS	IV
Radio subscription rate, %	−0.086* [0.045]	−0.347*** [0.095]	0.032* [0.017]	0.222* [0.117]
Region fixed effects	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes
Observations	857	855	853	851
F-statistic for the exclusion of the instrument		50.19		20.30
Panel C: Altonji-Elder-Taber tests				
Index of observables (predicted signal strength)	−0.037 [0.070]		−0.047 [0.049]	
Index of observables (predicted nonlinear signal strength)		0.014 [0.210]		−0.348* [0.200]
Population, fifth-order polynomial	Yes	Yes	Yes	Yes
Observations	958	958	918	918
R ²	0.432	0.432	0.573	0.576

The sign flips in 1933

- Left columns: change in the Nazi vote from May 1928 to September 1930. Negative in signal strength.
- Right columns: change from November 1932 to March 1933. Positive.
- Same districts, same measure of exposure, opposite sign. What changed is who controlled the content.
- Fixing the signal map at earlier years and re-running gives the same flip, so it is not the spread of radio.

Nazi radio, five weeks of it

- The same 8.2 pp increase in predicted subscription in 1933 → a **1 pp larger** rise in the Nazi vote between November 1932 and March 1933.
- With actual subscriptions rather than predicted ones: 2.0 pp.
- Counterfactual with the radio switched off in January 1933: 41.0% instead of 43.9%.
- With their coalition partner's 8%, that would not have been a majority of seats.
- About a quarter of what the NSDAP gained in those four months, from five weeks of broadcasting.

Persuasion rates, and why they differ

- “Do not vote for the extremist parties” in September 1930: **36.8%**.
- “Vote for the Nazis” in March 1933: **9.8%**.
- The 1933 campaign was five weeks, against a much longer one in 1930.
- It also targeted the uneducated working poor, the group that switched most in 1933 and owned radio sets least.
- Persuading someone not to vote for a party is easier than persuading them to vote for it: abstention counts.

TABLE V
RADIO AVAILABILITY AND VOTING IN ANTI-VERSAILLES TREATY REFERENDUM AND APRIL
1932 PRESIDENTIAL ELECTIONS

	(1)	(2)	(5)	(6)	(7)	(8)
	Referendum on the Law against the Enslavement of the German People		1932 presidential elections, 1st round			
	Voted “yes” in the referendum (share of eligible voters)		Von Hindenburg vote share		Hitler vote share	
Panel A: radio availability and other voting outcomes.						
Radio signal strength	−0.065** [0.031]		0.054** [0.026]		−0.049 [0.036]	
Radio signal strength, non-linear transformation		−0.244** [0.095]		0.194 [0.118]		−0.240 [0.149]
Nazi party vote share in 1928	0.647*** [0.090]	0.639*** [0.090]	−0.457*** [0.070]	−0.456*** [0.071]	0.584*** [0.064]	0.584*** [0.064]
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	949	949	952	952	952	952
R ²	0.845	0.846	0.881	0.881	0.876	0.876
Panel B: Altonji-Elder-Taber tests						
Index of observables (predicted signal strength)	−0.138 [0.133]		−0.101 [0.174]		−0.161 [0.157]	
Index of observables (predicted nonlinear transformation of signal strength)		−0.525 [0.393]		−1.291* [0.694]		−0.113 [0.634]
Nazi party vote share in 1928	1.276*** [0.091]	1.283*** [0.090]	−1.490*** [0.129]	−1.509*** [0.119]	1.490*** [0.128]	1.526*** [0.123]
Population, fifth-order polynomial	Yes	Yes	Yes	Yes	Yes	Yes
Observations	949	949	952	952	952	952
R ²	0.229	0.231	0.202	0.213	0.227	0.225

Other votes in the same direction

- The 1929 referendum against the Treaty of Versailles: radio programming was slanted against a yes.
- 8.2 pp more predicted subscription → 2 percentage points less support, an 11.4% fall.
- The 1932 presidential election: positive for the incumbent von Hindenburg, negative for Hitler, as predicted, but imprecise.
- Both outcomes control for the 1928 Nazi vote, to hold pre-existing political preferences fixed.

TABLE VI
DISCRIMINATION AND VIOLENCE AGAINST JEWS

	(1)	(2)	(3)
	Incidence of discrimination and violence against Jews		
Time span of the sample Specification	1929–1932 Panel	1933–1934 Pooled cross-section	1929–1934 Panel
Radio signal strength	−0.0020 [0.0019]	0.0020* [0.0011]	
Radio signal strength × pro-Nazi slant			0.0012** [0.0006]
Baseline controls, interacted with time fixed effects	Yes		Yes
City fixed effects	Yes		Yes
Time fixed effects	Yes	Yes	Yes
Region fixed effects, baseline controls		Yes	
Observations	5,328	2,664	7,992
Number of cities	1332	1332	1332
R^2	0.130	0.338	0.410

Notes. Standard errors clustered by city in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$. Pro-Nazi slant equals −1 between 1929 and 1932 and +1 between 1933 and 1934. Baseline controls include log(population), altitude, dummy for being located on a navigable river, share of Jewish population in 1925, share of Catholic population in 1925, share of blue-collar workers in 1925, share of white-collar workers in 1925, dummy for the data source, number of war participants per 1,000, number of welfare recipients per 1,000, number of pensioners with social assistance per 1,000, turnout and vote shares of DNVP, NSFP, SPD, and Zentrum in December 1924.

Discrimination and violence against Jews

- City-level records of verbal antisemitism, physical violence and property damage, 1929 to 1934.
- Negative with radio in 1929 to 1932, positive in 1933 to 1934, imprecisely estimated in both windows.
- The interaction with slant, in the last column, is the sharper statement: the sign follows the content.

TABLE VII
RADIO AVAILABILITY AND NAZI PARTY MEMBERSHIP

	(1)	(2)	(3)	(4)	(5)	(6)
Time period	Log of the number of new party members of NSDAP					
	Jan 1932–Jan 1933		Feb–May 1933			
Panel A: cross-sectional estimates						
Radio signal strength	0.0032 [0.0020]		0.0053** [0.0020]		0.0052** [0.0020]	
Radio signal strength, nonlinear transformation		0.0006 [0.0099]		0.0201** [0.0077]		0.0203** [0.0079]
Log of new party membership in 01/1932–01/1933					0.0544* [0.0292]	0.0579* [0.0289]
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	613	613	613	613	613	613
R ²	0.677	0.676	0.344	0.345	0.348	0.349
Panel B: Altonji-Elder-Taber tests						
Index of observables (predicted signal strength)	0.0026 [0.006]		0.0013 [0.004]		0.0016 [0.004]	
Index of observables (predicted nonlinear signal strength)		0.0221 [0.029]		0.0077 [0.017]		0.0080 [0.017]
Log of new party membership in 01/1932–01/1933					0.0484* [0.027]	0.0480* [0.027]
Population, fifth-order polynomial	Yes	Yes	Yes	Yes	Yes	Yes
Observations	613	613	613	613	613	613
R ²	0.587	0.587	0.288	0.288	0.292	0.292

Recruiting party members

- New NSDAP members by city, from the party membership cards.
- In January 1932 to January 1933, with pro-Weimar radio: nothing. Those who joined then were core supporters.
- In February to May 1933, after the takeover of the radio: a clear positive relationship with signal strength.
- The implied magnitude: **58% of new members** who joined in the first two months of Nazi control of the broadcast were persuaded by radio propaganda.
- The party stopped accepting members in May 1933 because of the flood of applicants.

TABLE VIII
RADIO AVAILABILITY AND ANTI-SEMITISM

Model	(1) Log(deportations before 1942) OLS	(2) Letters to <i>Der Stürmer</i> ML	(3) Attacks on synagogues OLS All cities with synagogues in 1933
Sample	All cities	All cities	
Panel A: Baseline results.			
Radio signal strength in 1937	0.0187** [0.0069]	0.0194*** [0.0068]	0.0001 [0.0014]
Log (population)	0.3447*** [0.0803]	0.3900*** [0.0634]	-0.0097 [0.0141]
Altitude	-0.0009* [0.0005]	-0.0008 [0.0005]	-0.0001 [0.0002]
City located at navigable river	0.3693*** [0.0683]	0.7504*** [0.1191]	0.0467** [0.0189]
Socioeconomic controls	Yes	Yes	Yes
Voting controls, 1924	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes
Observations	1,325	1,391	1,134
R^2	0.385		0.103
Panel B: Altonji-Elder-Taber style test			
Index of observables	-0.0084 [0.0263]	0.0245 [0.0221]	0.0103** [0.0045]
(Prediction of signal strength)			
Population	Yes	Yes	Yes
Geographic variables	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes
Observations	1,325	1,391	1,133
R^2	0.197		0.064

Notes. Results of OLS estimation in columns (1) and (3); results of negative binomial maximum likelihood estimation in column (2). Standard errors are clustered by region (Wahlkreis). * $p < .1$, ** $p < .05$, *** $p < .01$. Unit of observation is city in Voigtländer and Voth (2012) sample. Socioeconomic controls include share of Jewish and Catholic population, shares of blue- and white-collar workers in 1925, number of war participants per 1,000, number of welfare recipients per 1,000, number of renters of social housing per 1,000, log of average property. Voting controls include voter turnout and vote shares of DNVP, NSFP, SPD, and Zentrum in December 1924. Geographic controls include altitude and dummy for being located on a navigable river.

After power was fully consolidated

- Elections stop being informative in a dictatorship, so the outcomes change.
- Jews deported to concentration camps before 1942, mostly following denunciations by neighbors and local officials.
- Open antisemitic letters to the Nazi newspaper *Der Stürmer*, 1935 to 1938. Attacks on synagogues in November 1938.
- Stronger signal in 1937 (one standard deviation) → **21% more deportations** (mean 11.7 per city) and **22% more letters** (mean 1.76, so 0.38 more).
- No effect on synagogue attacks.

TABLE IX

RADIO AVAILABILITY AND ANTI-SEMITISM, THE ROLE OF HISTORIC PREDISPOSITIONS

Model	(1) Log(deportations before 1942) OLS	(2) Letters to <i>Der Stürmer</i> ML	(3) Attacks on synagogues OLS
Panel A: interactions with pogroms			
Sample	Subsample of cities with historical Jewish settlement		
Pogroms in 1349 * Radio signal strength, demeaned	0.048*** [0.016]	0.073*** [0.016]	0.003 [0.007]
Radio signal strength, 1937	-0.032* [0.017]	-0.058*** [0.016]	-0.004 [0.007]
Pogroms in 1349	0.701*** [0.171]	0.601*** [0.227]	0.160** [0.065]
All baseline controls	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes
Observations	296	319	274
R^2	0.658		0.253

Propaganda meets prior beliefs

- Cities are split by historical antisemitism: whether they had anti-Jewish pogroms during the Black Death of 1349.
- Where the pogroms happened, the effect of radio is 2.6 to 3.8 times larger than in the average city.
- Same pattern using the 1924 vote for the extreme nationalist NSFP, and using land inequality in 1895.
- **Where they did not happen, the sign flips:** 28% fewer deportations and 45% fewer letters.

Persuasion or coordination?

- Propaganda could work as persuasion, or merely as coordination: a signal that certain acts will not be punished.
- Pure coordination cannot produce a *negative* effect anywhere.
- So at least part of what radio did was persuasion, or dissuasion, with the sign set by prior beliefs.
- “it can backfire, if listeners are unlikely to believe its message”.
- Listeners who hear claims they know to be false update against the regime.

What the paper documents

- Mass media were a safeguard against the fall of an unconsolidated democracy, and then a tool in its collapse.
- What flipped the sign was control over content, plus whether extremists were barred from the air.
- Under the dictatorship, radio raised popular participation in the regime's worst policies.
- But only where the audience did not disagree a priori. Elsewhere it produced resistance.

Historical Antisemitism, Ethnic Specialization, and Financial Development

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Historically, European Jews have specialized in financial services while being the victims of antisemitism. We find that the present-day demand for finance is lower in German counties where historical antisemitism was higher, compared to otherwise similar counties. Households in counties with high historical antisemitism have similar saving rates but invest less in stocks, hold lower saving deposits, and are less likely to get a mortgage to finance homeownership after controlling for wealth and a rich set of current and historical covariates. Present-day antisemitism and supply-side forces do not fully explain the results. Households in counties where historical antisemitism was higher distrust the financial sector more—a potential cultural externality of historical antisemitism that reduces wealth accumulation in the long run.

Key words: Cultural economics, Cultural finance, Intergenerational transmission of norms, Stereotypes, Household finance, History & finance.

JEL Codes: D91, G11, J15, N90, Z10, Z12

Related work: historical antisemitism today

- D'Acunto, Prokopczuk and Weber, on the same historical measure of German antisemitism.
- German counties where historical antisemitism was higher show lower present-day demand for finance.
- Households there save at similar rates but hold less in stocks and in deposits, and distrust the financial sector more.
- The 1349 pogrom variable in Adena et al. is doing work in a literature well beyond propaganda.

A tear in the iron curtain

Bursztyn and Cantoni (2016)

The question

- Autocratic regimes restrict not only what citizens can choose, but what they can know.
- Censorship is never complete: the Great Firewall, Radio Free Europe, jammed and unjammed frequencies.
- “Does the information that passes these cracks matter?”
- The outcome here is not voting or violence. It is **private consumption**.

The setting

- East Germany, where most households could watch West German television and some could not.
- The regions without reception were the northeastern and southeastern corners: too far from the border and West Berlin, or behind mountains.
- Together, about one tenth of the East German population.
- Dresden, in the Elbe valley, became known as the “valley of the clueless”.

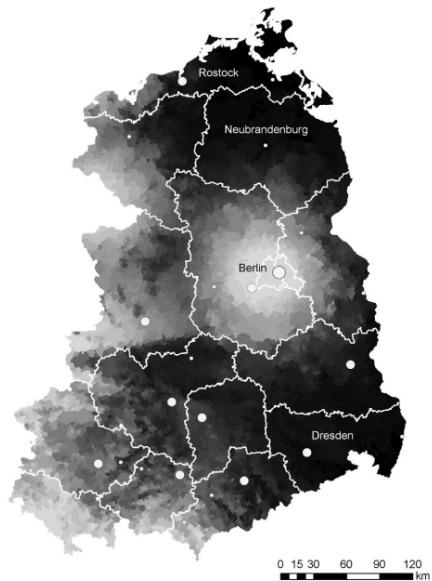
Television in the GDR

- ARD in the West and DFF in the East both started broadcasting in 1952. Over 300,000 sets in the GDR by the end of 1958.
- By 1989, an estimated 98% of East German households had a TV set and 46% a color one.
- By 1988 one household in six had more than one set: 117 sets per 100 households.
- East German television was tightly controlled by the party, with “a drab mixture of political propaganda and Soviet-produced movies”.
- **No advertising on East German television** until shortly before reunification.

The regime largely let it happen

- Authorities mostly closed an eye on the antennas needed to pick up West German channels.
- The West German television frequencies were not jammed, though this was feasible and was done to radio.
- The 1985 youth survey: respondents watched on average more than two hours of West German television each weekday.
- Survey and archival evidence says they watched it mainly for entertainment shows, and the advertisements in them.

FIGURE 1.—SIGNAL STRENGTH (ARD) IN EAST GERMANY, 1989

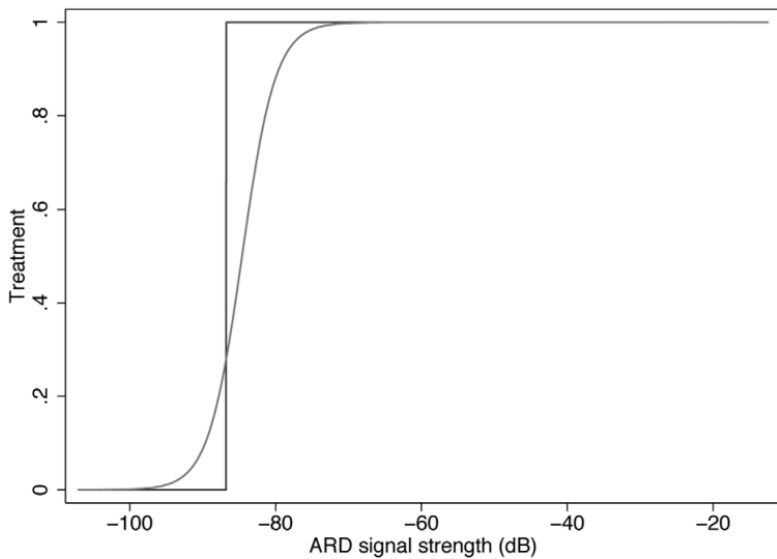


Dark: weak signal. Dots represent major cities. District borders are (*Bezirke*) superimposed as white lines.

Reading the signal map

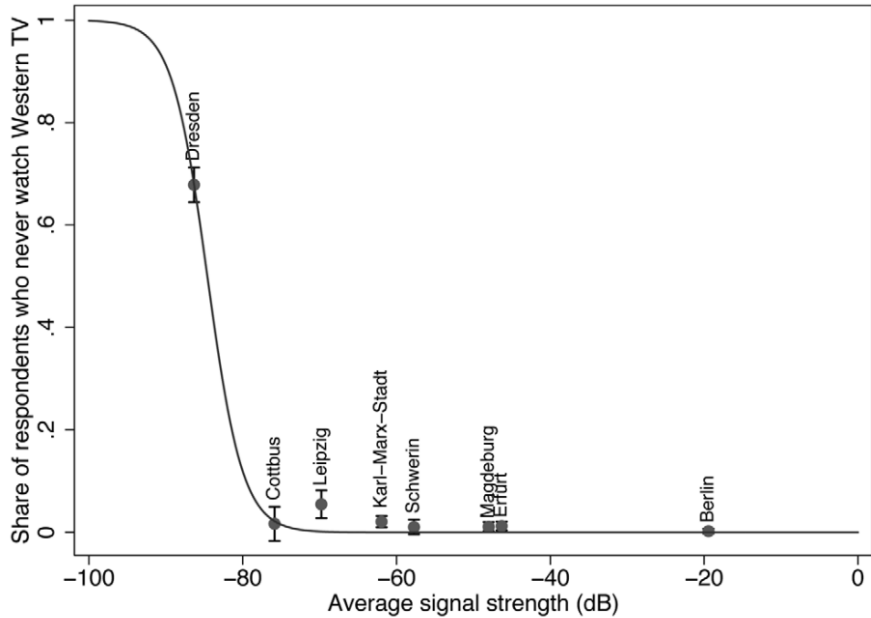
- Predicted ARD signal strength in 1989, dark meaning weak.
- Bright around West Berlin and along the western border, dark in the northeast and around Dresden.
- Antennas, heights, power and frequencies from 1989, run through the Irregular Terrain Model on a 1 kilometer grid.
- Values run from -107 dB in Sassnitz on the island of Rügen to -10.9 dB in Seeburg, on the West Berlin border.

FIGURE 4.—TREATMENT DEFINITIONS



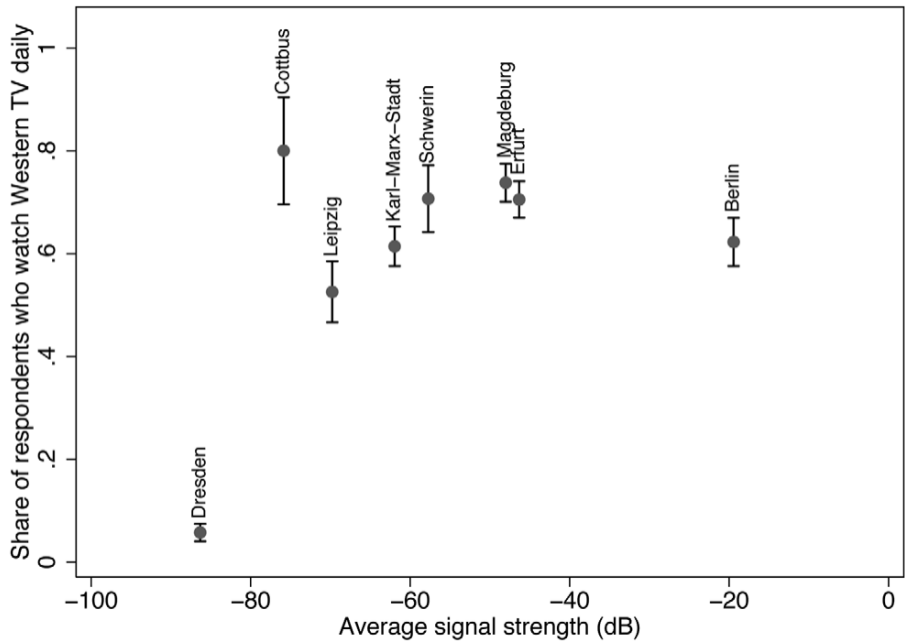
Defining treatment

- Reception does not fade smoothly: below a threshold the noise wins and nothing comes through.
- The threshold is anchored on Dresden, whose city average was -86.8 dB. At or below that, a municipality is control.
- Control is built to contain only households that with certainty had no access.
- The smooth curve is the alternative: a logistic fit to actual 1985 viewership, read as the probability of watching.



Never watching, by district

- An anonymous 1985 survey of East German youths, asked how often they watched Western channels.
- Cottbus, average signal -75.9 dB: 1.67% say they never watch.
- Dresden, average signal -86.3 dB: 67.85%.
- Everywhere with a stronger signal sits on the floor. **The step is where the model says it should be.**



Watching daily, by district

- The mirror image: 66.28% in districts with access say they watch Western television every day.
- In Dresden, 5.72%.
- Berlin sits slightly lower than the rest despite the strongest signal.
- These are stated answers under an authoritarian regime, but the survey was anonymous and is generally treated as valid.

Why these regions can be compared

- Access is set by distance, terrain and where the West German antennas happened to stand.
- Both areas contain industrial districts (Dresden in control, Leipzig and Halle in treatment) and agricultural ones.
- Residential mobility was tightly limited in a planned economy, so sorting into reception areas is implausible.
- The concern to rule out: that goods simply arrived later in the remote east.

TABLE 1.—REGIONAL CHARACTERISTICS, 1955 AND 1990 (DISTRICT LEVEL), BY TREATMENT STATUS

	Treatment	Control	Difference	SE	P-Value
<i>A: 1955 (District-Level Data)</i>					
Population density (inhabitants/km ²)	206	202	4	77	0.959
Share of employed in agriculture (%)	23.7	27.8	-4.1	11.1	0.744
Share of employed in industry (%)	34.1	28.7	5.4	10.0	0.635
Retail sales per capita (mark)	1,691	1,694	-3	102	0.979
Savings per capita (mark)	277	297	-20	28	0.544
<i>B: 1990</i>					
Population density (inhabitants/km ²)	181	176	5	62	0.941
Share of employed in agriculture (%)	13.5	11.3	2.2	5.1	0.706
Share of employed in industry (%)	33.2	39.5	-6.3	7.5	0.479
Retail sales per capital (mark)	7,577	7,250	327	188	0.190
Savings per capital (mark)	9,312	9,381	-69	928	0.946
Cars per 1,000 inhabitants	237.4	237.6	-0.2	12.1	0.992
<i>C: Difference 1990–1955</i>					
Population density (inhabitants/km ²)	-18	-26.2	8.2	15.4	0.626
Share of employed in agriculture (%)	-14.5	-12.6	-1.9	6.0	0.778
Share of employed in industry (%)	5	5.5	-0.5	3.0	0.870
Retail sales per capital (mark)	5,862	5,557	305	157	0.142
Savings per capital (mark)	8,946	8,994	-48	770	0.954

Population-weighted averages, excluding the district of East Berlin. *P*-values based on weighted Welch's *t*-tests of difference in means (two-sided, allowing for unequal variances). Total number of districts: fourteen (eleven treatment, three control).

Source: Statistisches Amt der DDR (1955, 1990).

Districts in 1955 and 1990

- Population density, employment in agriculture and industry, retail sales and savings per head.
- In 1955, before Western television mattered, the two groups are hard to tell apart.
- In 1990, at the end of the GDR, the same. Cars per 1,000 inhabitants: 237.4 against 237.6.
- The bottom panel compares the changes between the two years: no differential trend.

TABLE 2.—MIGRATION RATES, 1991–1993 (COUNTY LEVEL), BY TREATMENT STATUS

Variable	Treatment	Control	Difference	SE	<i>P</i> -Value
Migration rate to West Germany (%), 1991	0.368	0.343	0.024	0.034	0.480
Migration rate to West Germany (%), 1992	0.572	0.566	0.006	0.028	0.845
Migration rate to West Germany (%), 1993	0.596	0.607	−0.011	0.029	0.704

Total number of counties (*Kreise*): 215 (192 treatment, 23 control). *P*-values based on *t*-tests of difference in means, allowing for unequal variances.

Source: Own analysis based on migration matrices (by *Kreis*), German Federal Statistical Office.

TABLE 3.—SUPPLY SIDE: DIFFERENCES ACROSS TREATMENT REGIONS, 1993 (COUNTY LEVEL)

Variable	Treatment	Control	Difference	SE	<i>P</i> -Value
Food supermarkets	0.075	0.070	0.004	0.015	0.782
Mail-order companies	0.011	0.011	−0.001	0.002	0.844
Other retail stores	1.767	1.715	0.052	0.083	0.544

Variables are expressed as number of plants per 1,000 inhabitants. Total number of counties (*Kreise*): 86 (75 treatment, 11 control). *P*-values based on *t*-tests of difference in means, allowing for unequal variances.

Source: Own analysis based on the IAB Establishment History Panel (BHP).

Moving away, and what was in the shops

- Migration to West Germany in 1991, 1992 and 1993: the same in treatment and control.
- So the people observed after reunification are not a selected remainder.
- Establishment data for 1993: food supermarkets, mail-order companies and other retail stores per 1,000 inhabitants.
- No difference in local supply, which is the alternative story for any consumption gap.

TABLE 4.—BALANCEDNESS OF COVARIATES, EVS

Variable	Treatment	Control	Difference	SE	P-Value
<i>EVS 1993</i>					
Disposable income	41,268.8	41,676.4	-407.6	1,028.6	0.692
Age	50.1	47.9	2.2	1	0.028
Number of children	0.659	0.787	-0.127	0.045	0.005
Single	0.282	0.246	0.036	0.023	0.125
Female	0.422	0.43	-0.008	0.025	0.759
German	0.997	0.998	-0.002	0.002	0.332
Employed	0.517	0.498	0.019	0.025	0.446
Retired	0.274	0.248	0.026	0.024	0.282
On Welfare	0.025	0.027	-0.003	0.009	0.752
Small city (< 5000 inhabitants)	0.214	0.19	0.025	0.021	0.230
<i>EVS 1998</i>					
Disposable income	49,807.8	48,201.0	1,606.8	1,106.6	0.147
Age	51.7	52.7	-1.1	0.8	0.161
Number of children	0.409	0.347	0.062	0.021	0.004
Single	0.303	0.358	-0.055	0.021	0.008
Female	0.425	0.438	-0.014	0.019	0.486
German	0.995	0.999	-0.003	0.001	0.007
Employed	0.552	0.530	0.023	0.019	0.247
Retired	0.346	0.363	-0.018	0.02	0.366
On welfare	0.016	0.009	0.006	0.004	0.065
Small city (< 5000 inhabitants)	0.237	0.174	0.063	0.013	0.000

Total number of observations: 9,305 in 1993 (8,035 treatment, 1,270 control) and 11,794 in 1998 (10,147 treatment, 1,647 control). *P*-values based on *t*-tests of difference in weighted means (survey weights used).

The households in the survey

- German income and expenditure survey (EVS), waves of 1993 and 1998.
- A repeated cross-section: the same household is not followed over time.
- The restricted version identifies the municipality, which is what allows linking each household to reception.
- Treatment and control households look alike on income, employment, welfare receipt and city size; treated households are slightly older with fewer children.

Why look after 1990 and not before

- Before reunification, wanting a Western good could not show up as buying it.
- Consumption in the GDR was regimented by the Ministry of Commerce and Provisioning, and set by what was in the store that day.
- After reunification anything seen on television could in principle be bought.
- West German broadcasts reached all of the former GDR by 1990, so the treatment stops right when the outcome starts.

No effect on the aggregates

- Disposable income, total private consumption and savings: nothing, in 1993 and in 1998.
- The 1993 estimate for total private consumption is -0.83% , and rules out an increase above 1.5% .
- Consumer credit and overdrafts: nothing either. The 1993 estimates rule out a propensity to use consumer credit 6% higher, and overdrafts 1% higher.
- So three decades of Western television did not make East Germans spend or borrow more overall.

The advertising channel

- Advertising minutes per day by category on ARD, averaged over 1980 to 1989.
- Total advertising was small: under 20 minutes a day, regulated by law, mostly between 7 and 8 in the evening.
- Categories with heavy advertising should be where treatment and control differ, if television did anything.
- East Germans were in no sense the intended audience of those commercials.

TABLE 7.—MOST AND LEAST ADVERTISED CATEGORIES OF GOODS

Goods	Average Minutes of Advertising per Day	Share of Advertising Time (%)	Average Budget Share, 1993 (%)
Food and drinks	6'49"	42.5	23.7
Body and mouth care	2'52"	17.9	3.4
Cleaning and washing	1'46"	11	0.7
Media, leisure	0'53"	5.47	6.3
Home and garden	0'35"	3.66	11.7
Pharmaceuticals	0'31"	3.19	0.7
Cars, accessories	0'28"	2.95	14.1
Transportation and tourism	0'11"	1.15	5.9
Photo, optics, watches, jewelry	0'8"	0.83	1.1
Clothes, textiles	0'8"	0.78	8.4
Other	0'0"	0	24.1

Figures correspond to advertising time allocated to each of the categories of goods on the main West German state TV channel (ARD); average of the years 1980 to 1989.

Source: Zentrallausschuss der Werbewirtschaft (1981–1990).

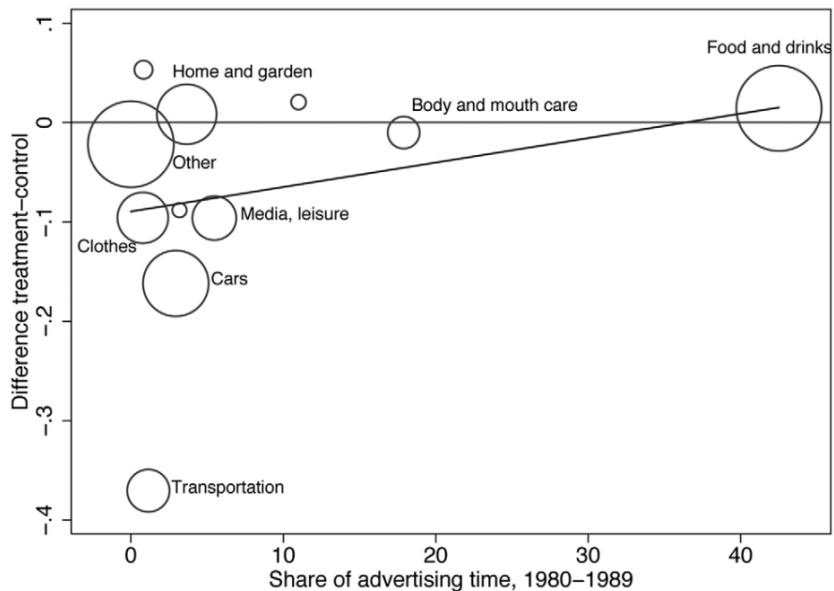
Which goods were advertised

- Food and drinks: 6 minutes 49 seconds a day, **42.5%** of all advertising time.
- Body and mouth care: 2 minutes 52 seconds, 17.9%.
- Clothes and textiles: 8 seconds, 0.78%, against a budget share of 8.4%.
- Budget shares and advertising shares are far apart, which is what makes the test informative.
- An eleventh category collects everything never advertised: rent, utilities, bicycles, telephones.

How to read the next two figures

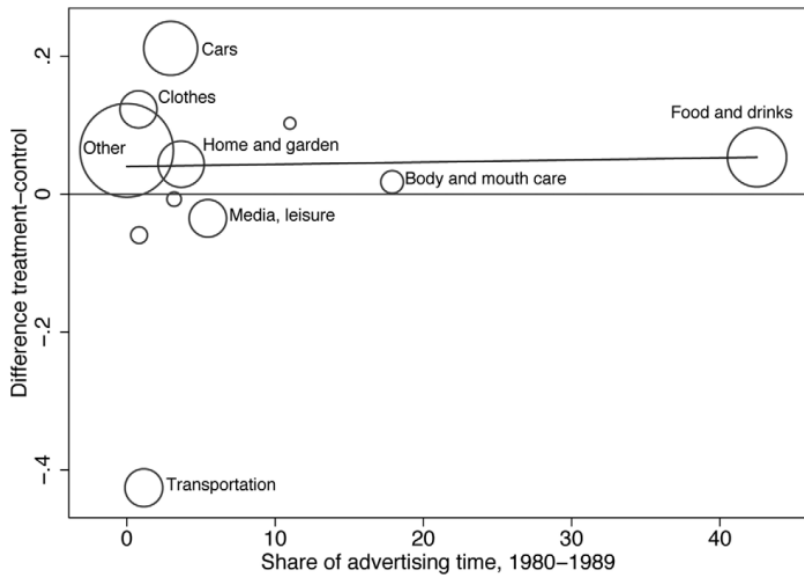
- Horizontal axis: the category's share of total ARD advertising time, 1980 to 1989.
- Vertical axis: the raw difference between treatment and control in average yearly spending, in logs.
- Circle size: the category's share of the average household budget.
- The fitted line weights categories by budget share, so tiny categories do not drive it.

FIGURE 5.—DIFFERENCES IN CONSUMPTION LEVELS AND ADVERTISING, 1993



- Most categories sit below zero, consistent with the slightly negative effect on total consumption.
- The line slopes up: the more a category was advertised, the smaller the gap, and for food and drinks it turns positive.
- Transportation and tourism, almost never advertised, sits far below.
- This is a recomposition of spending, not more spending.

FIGURE 6.—DIFFERENCES IN CONSUMPTION LEVELS AND ADVERTISING, 1998



1998

- Same axes, five years later. The line is essentially flat.
- Cars and clothes, both lightly advertised in the 1980s, now sit above the line.
- Whatever the old advertisements did, it is gone.

The size of the recomposition

- In 1993, spending on a category with one more minute of daily advertising was about **1.5% higher** in the areas that could watch.
- Per percentage point of total advertising time, about 0.25% more spending.
- Body and mouth care, at 17.9% of advertising time, works out to roughly 4.4% more.
- Categories with no advertising at all: consumption almost 9% lower in the treated areas.
- By 1998 the estimate is small and no longer distinguishable from zero.

Robustness

- Including East Berlin, adding household characteristics, or using the continuous treatment: unchanged.
- Moving the threshold to -84.8 , -82.8 and -80.8 dB: unchanged.
- Restricting to regions within 30 dB and 20 dB of the threshold: unchanged.
- Within 10 dB it disappears, and the paper explains why: there the comparison is no exposure against very little exposure.
- Controlling for driving distance to West Germany, and for its interaction with advertising, leaves the result.

What the paper documents

- Thirty years of Western television did not raise total consumption, savings or borrowing in East Germany.
- It shifted the *composition* of spending toward the goods that had been advertised.
- Advertising does more than move consumers between brands: it moves them between broad categories.
- The effect is a memory of the 1980s, and it fades within a decade.

Taking stock

Three settings, one source of variation

- All three read the same physics: transmitter power, distance, terrain, then the Irregular Terrain Model.
- Croatia: reception varies village by village, in a jagged way, near the Serbian border.
- Germany in the 1930s: reception varies across districts and over time as transmitters open.
- East Germany: reception varies across a threshold set by valleys and distance.
- In each case the listener did not choose the signal, and often was not the intended audience.

What determines whether media move people

- **Content**: the same German radio, in the same districts, pushed against the Nazis and then for them.
- **Control**: who holds the transmitters, and whether extremists are allowed on the air.
- **Predisposition**: propaganda worked where the audience half believed it and backfired where it did not.
- Croatia adds a fourth: the audience does not have to be the target, and entertainment can carry the politics.

How big are these effects

- Serbian radio in Croatia: a persuasion rate of 4.3%, on the low end of the literature.
- Nazi radio in the March 1933 campaign: 9.8%.
- Weimar radio in 1930, telling people not to vote for extremists: 36.8%.
- Small shares of the audience, but enough to change which coalition can form a majority.

Where else this shows up

- Yanagizawa-Drott: hate speech on the radio and casualties in the Rwandan genocide of 1994.
- Paluck and Green: an appeasing educational broadcast in post-conflict Rwanda, run as a field experiment.
- Gentzkow and Shapiro: media in Arabic countries and anti-American sentiment.
- La Ferrara, Chong and Duryea: exposure to television soap operas and fertility in Brazil.
- For discussion: in Brazil, what would play the role of the terrain here?